

Trainings ▾

Preparatory Steps

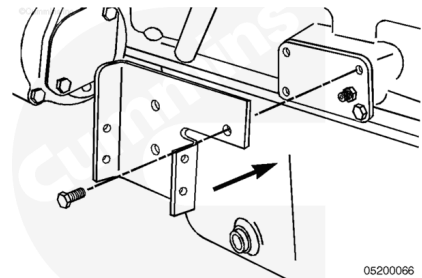
Remove the oil control valve, if applicable. Refer to Procedure 007-078 (</qs3/pubsys2/xml/en/procedures/96/96-007-078-tr.html>).



Remove

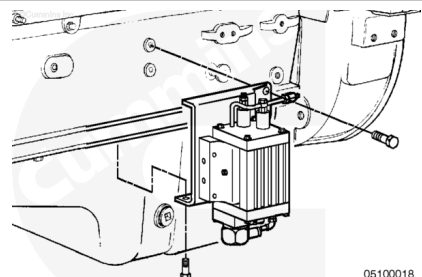
L10 and M11 Engines

Remove the oil control valve mounting bracket.



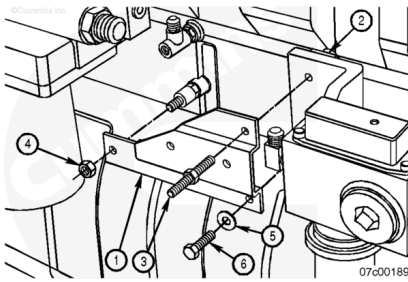
N14 Engines

Remove the oil control valve mounting bracket.



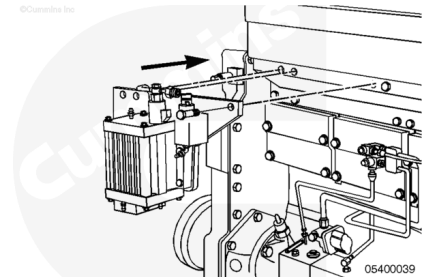
ISX Engines

Remove the oil control valve mounting bracket.



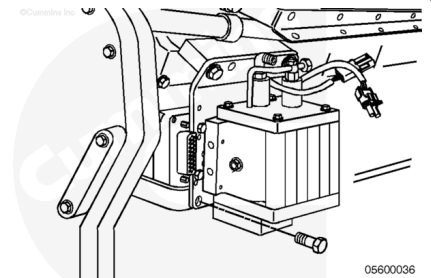
K19 Engines

Remove the oil control valve and oil control valve mounting bracket as an assembly.



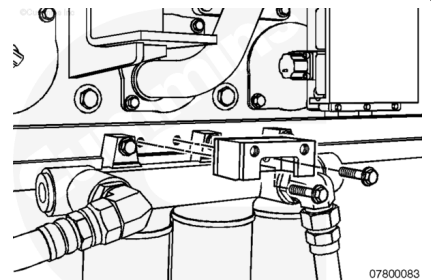
K38 and K50 Engines

Remove the oil control valve and oil control valve mounting bracket as an assembly.



QSK45 and QSK60

Remove the oil control valve mounting bracket.



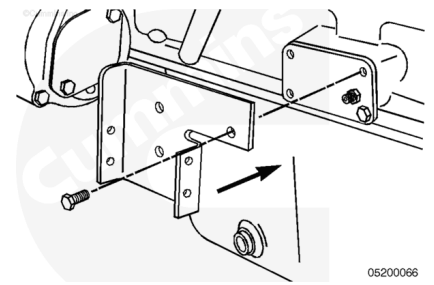
Install

L10 and M11 Engines

Install the oil control valve mounting bracket using the M10 X 1.5 x 25-mm bolts.

Tighten the four M10 bolts that are used to secure the mounting bracket and handhole cover.

Torque Value: 45 n•m [33 ft-lb]



N14 Engines

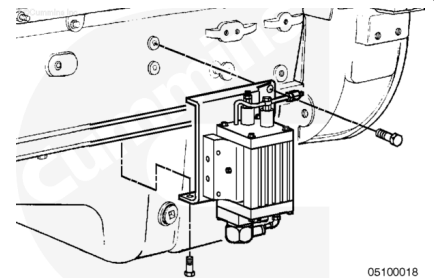
Note : Install all four oil control valve mounting bracket bolts prior to tightening.

Install two 3/8-24 X 3/4 bolts into the top two bolt holes of the oil control valve bracket.

Torque Value: 40 n•m [30 ft-lb]

Install the two 7/16-20 X 1.375 bolts into the bottom of the oil control valve mounting bracket.

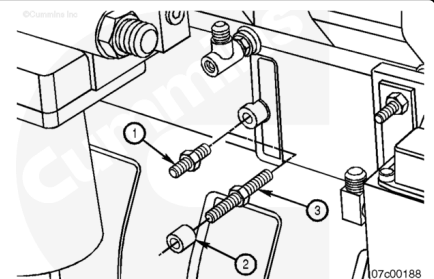
Torque Value: 55 n•m [41 ft-lb]



ISX With Lift Pump

Install the stud (3) where the previous stud was removed.

Install a spacer (2) over the top of the newly installed stud.



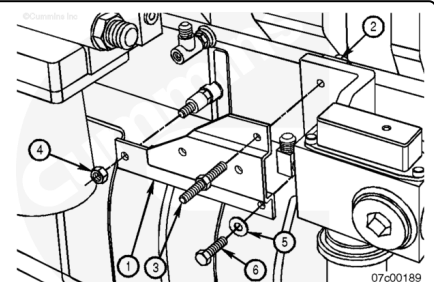
Place the Centinel™ bracket (1) over the stud with the spacer that was previously installed and line up the two right-hand holes with the holes in the lift pump.

Place the wear plate (2) retained in the previous step between the lift pump and the block. Install the stud (3) through the top hole of the Centinel™ bracket, lift pump and wear plate into the block. Hand tighten the stud.

Install a nut (4) over the left stud until hand tight. Place the M10 flat washer (5) over a M10-1.5 x 40 mm capscrew (6) and install the capscrew into the bottom right-hand hole.

Tighten all three fasteners.

Torque Value: 47 n•m [35 ft-lb]



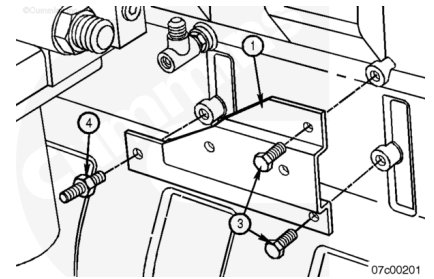
ISX Without Lift Pump

Install the bracket (1) with the stud (4) to the cylinder block mounting hole. Line up the two right-hand holes with the mounting holes in the cylinder block.

Install two M10-1.5 x 20 mm capscrews (3) through the two holes of the Centinel™ bracket and wear plate into the block.

Tighten all three fasteners.

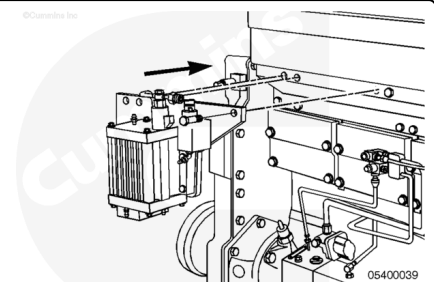
Torque Value: 47 n·m [35 ft-lb]



K19 Engines

Using the three 3/8-16 x 7.25 bolts provided, install the oil control valve bracket onto the engine.

Torque Value: 40 n·m [30 ft-lb]



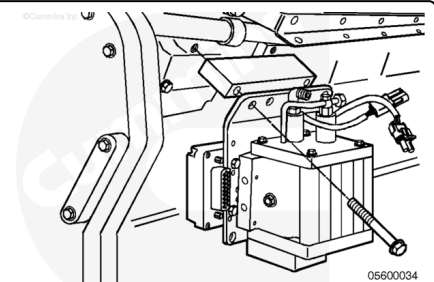
K38 and K50 Engines

Locate the spacer in the upfit kit.

Note : There are two valve bracket to engine mounting bolt sizes, depending on the type of aftercooler that the engine is configured with. Use the step that applies to your particular aftercooler type.

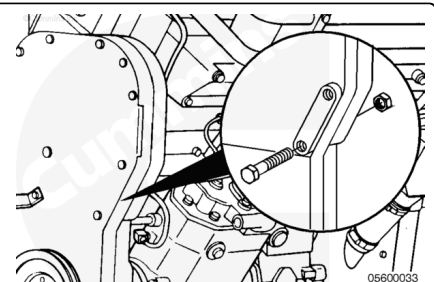
- For low-temperature aftercooled engines, install the oil control valve bracket onto the engine with the top two bolt holes using the two 3/8-16 x 7 bolts provided and the spacer.
- For aftercooled engines, install the oil control valve bracket onto the engine with the top two bolt holes using the two 3/8-16 x 7.50 bolts provided and the spacer.

Torque Value: 40 n·m [30 ft-lb]



Install the 7/16-14 x 5 extended thread bolt provided to where the bolt was just removed.

Torque Value: 60 n·m [44 ft-lb]

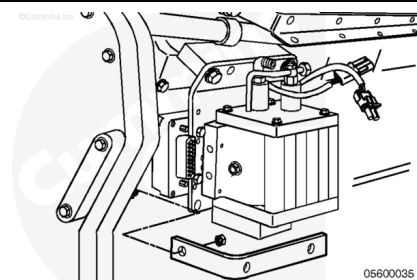


Locate the valve bracket bottom support bracket in the upfit kit.

Install the support bracket onto the 7/16-14 bolt.

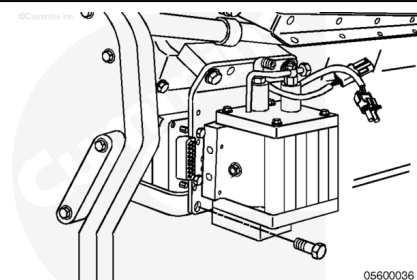
Install a 7/16-14 nut onto the bolt.

Torque Value: 60 n•m [44 ft-lb]



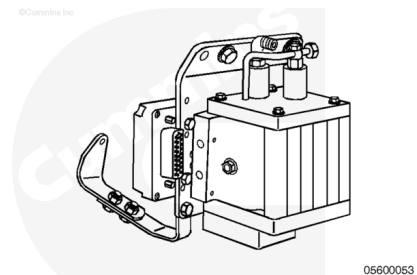
Bolt the support bracket to the bottom of the valve bracket using two 3/8-16 x 1.25 bolts, washers, lock washers, and nuts supplied.

Torque Value: 40 n•m [30 ft-lb]



Note : There is a supplemental mounting bracket for the K38/50 installation.

It is used to install the Centinel™ above the number 4 cylinder handhole cover in the event that the air compressor mounted on the engine is in the way.



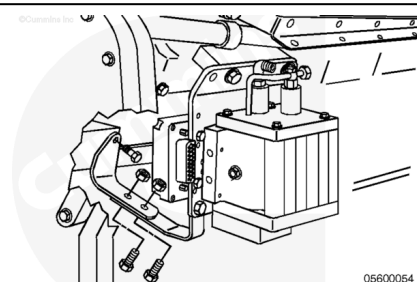
The installation will use the same mounting hardware with the exception of the lower brackets.

The optional brackets are a two-piece design to allow for easier installation.

There are four extra nuts and bolts for securing the length of the optional brackets.

Adjust the brackets so that the control valve will be vertical to the engine and tighten the bolts.

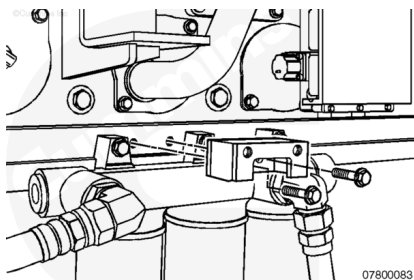
Torque Value: 40 n•m [30 ft-lb]



QSK45 and QSK60

Install the mounting bracket using the two M12 x 1.75-35 capscrews onto the QSKV engine pan rail.

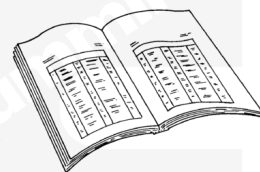
Torque Value: 65 n•m [48 ft-lb]



Finishing Steps

Install the oil control valve, if applicable. Refer to Procedure 007-078 (</qs3/pubsys2/xml/en/procedures/96/96-007-078-tr.html>).

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